ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

ROBOTICS FUNDAMENTALS

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 2. Section A: objective questions (30) · Section B: what robots are and where they are used, robot components, sensors and actuators, robot movement and configurations, safety, ethics and the future of work (70).

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INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL calculations.
5. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 A robot is best described as a machine that …

A can only be controlled by hand.
B has no moving parts.
C only displays information.
D can sense, process information and act, often automatically.

1.1.2 The "brain" of a robot that runs its program is the …

A gripper. B controller. C chassis. D battery.

1.1.3 The tool at the end of a robot arm, such as a gripper or welding torch, is the …

A base.
B encoder.
C power supply.
D end effector.

1.1.4 Each independent way in which a robot joint can move is called a …

A work envelope.
B degree of freedom.
C payload.
D cycle time.

1.1.5 The maximum mass a robot arm can carry safely is its …

A reach. B speed. C payload. D accuracy.

1.1.6 A robot that cleans floors by itself in a home is an example of a/an …

A service robot.
B industrial welding robot.
C exoskeleton.
D remote-controlled toy only.

1.1.7 Which sensor would a robot use to measure the distance to a wall?

A Temperature sensor
B Ultrasonic sensor
C Microphone only
D Limit switch only

1.1.8 A motor that can be moved to a precise angle, often 0° to 180°, is a …

A AC fan motor. B solenoid. C generator. D servo motor.

1.1.9 The word "robot" was first used in a play written by …

A Karel Čapek. B Isaac Newton. C Nelson Mandela. D Thomas Edison.

1.1.10 The total space that a robot arm can reach is its …

A base plate. B axis. C work envelope. D footprint only.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 All robots look like humans.

1.2.2 Sensors give the robot information about its environment.

1.2.3 Actuators are the parts of the robot that cause movement.

1.2.4 Industrial robots are often used for dangerous, dirty or repetitive jobs.

1.2.5 A remote-controlled car that has no sensors or program is an autonomous robot.

1.2.6 An emergency stop button must be easy to reach.

1.2.7 Pneumatic actuators use compressed air.

1.2.8 A drone can be used to inspect power lines.

1.2.9 Robots never make mistakes, so they never need maintenance.

1.2.10 A collaborative robot (cobot) is designed to work safely next to people.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Robot that works without human control, making its own decisions	A	autonomous robot
1.3.2	Wearable robotic frame that helps a person to lift or walk	B	drone
1.3.3	Unmanned flying robot	C	exoskeleton
1.3.4	Robot arm with rotating joints, similar to a human arm	D	articulated robot
1.3.5	Sensor that counts the rotations of a wheel or motor shaft	E	repeatability
1.3.6	Switch that is pressed when a robot part reaches the end of its movement	F	feedback
1.3.7	The frame or body on which the robot's parts are mounted	G	chassis
1.3.8	Using signals from sensors to correct the robot's actions	H	solenoid
1.3.9	Device that turns an electrical signal into a short straight push or pull	I	limit switch
1.3.10	Ability of a robot to return to the same position every time	J	encoder

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: ROBOTS AND THEIR USES

2.1 Explain the difference between a robot and an ordinary machine such as a washing machine. (3)

2.2 Name FIVE industries in South Africa where robots are used and give an example of a task in EACH. (10)

2.3 Give FOUR advantages and THREE disadvantages of using robots in a car factory. (7)

[20]

QUESTION 3: ROBOT COMPONENTS, SENSORS AND ACTUATORS

3.1 Name the FIVE main components of a robot system and state the function of EACH. (10)

3.2 A line-following robot must follow a black line on a white floor and stop if something is in front of it. Name TWO sensors it needs and explain how each one works. (6)

3.3 Compare a DC motor, a servo motor and a stepper motor by giving ONE typical use of each in a robot. (3)

3.4 Explain the difference between an open-loop and a closed-loop (feedback) control system, using a robot example. (4)

[23]

QUESTION 4: MOVEMENT AND CONFIGURATIONS

4.1 Name THREE types of robot arm configurations and give ONE use of each. (6)

4.2 A human arm can be compared with a robot arm. Match the shoulder, elbow, wrist and hand to the parts of a robot arm. (4)

4.3 A mobile robot's wheel has a diameter of 6,5 cm. Calculate how far the robot moves when the wheel turns 10 times. (Use π = 3,14.) (3)

4.4 Give TWO types of locomotion (ways of moving) for mobile robots, other than wheels. (2)

[15]

QUESTION 5: SAFETY, ETHICS AND THE FUTURE OF WORK

5.1 Give FOUR safety measures used to protect workers around industrial robots. (4)

5.2 Many people fear that robots will take their jobs. Discuss this statement, giving TWO arguments for and TWO against. (4)

5.3 Name FOUR new careers or job opportunities created by robotics. (4)

[12]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER